

# MYSQL PROJECT

**TOPIC: GYM MEMBERSHIP DATA ANALYSIS USING MYSQL**

**SOURCE: KAGGLE**

## INTRODUCTION

Gym Membership Data contains information about gym members such as their age, gender, membership type, weekly visitors, attendance days, fitness activities, sauna usage, and favorite drinks.

The main objective of this project is to analyze the data using MySQL queries and gain useful insights about member behavior, membership plans, and gym usage patterns.

**TOOL: MYSQL Workbench**

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## Queries

1. Find the total number of gym members.

```
SELECT COUNT(*)
```

```
FROM gym.gym_membership;
```



The screenshot shows the MySQL Workbench interface. At the top, there are tabs for 'Result Grid', 'Filter Rows', 'Export', and 'Wrap Cell Content'. Below these, a table with one row is displayed. The row has two columns: 'count(\*)' and '1000'. The 'count(\*)' column is highlighted in blue.

| count(*) |
|----------|
| 1000     |

**Insight: This query shows the total number of members registered in the gym.**

## Gym Membership Data Analysis using MYSQL

2.How many male and female members are there.

```
SELECT gender, COUNT(*)  
FROM gym.gym_membership  
GROUP BY gender;
```



| gender | count(*) |
|--------|----------|
| Female | 503      |
| Male   | 497      |

**Insight: This helps compare the gender distribution of gym members.**

3.Find the average age of members.

```
SELECT AVG(age)  
FROM gym.gym_membership;
```



| avg(age) |
|----------|
| 30.6040  |

**Insight: This shows the average age of people using the gym.**

4.Display only 10 members whose age is above the average age.

```
SELECT age, gender  
FROM gym.gym_membership  
WHERE age > (SELECT AVG(age) FROM gym.gym_membership)  
LIMIT 10;
```



| age | gender |
|-----|--------|
| 47  | Female |
| 41  | Male   |
| 44  | Male   |
| 44  | Male   |
| 46  | Male   |
| 41  | Male   |
| 35  | Female |
| 35  | Female |
| 36  | Female |
| 37  | Male   |

**Insight: This identifies members who are older than the average.**

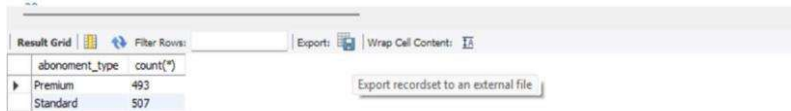
## Gym Membership Data Analysis using MYSQL

5.Count the number of members in each membership type.

```
SELECT abonoment_type, COUNT(*)
```

```
FROM gym.gym_membership
```

```
GROUP BY abonoment_type;
```



The screenshot shows a database interface with a 'Result Grid' tab. The grid contains two columns: 'abonoment\_type' and 'count(\*)'. There are two rows: 'Premium' with a count of 493, and 'Standard' with a count of 507. An 'Export' button and a 'Wrap Cell Content' checkbox are visible at the top right. A button labeled 'Export recordset to an external file' is located below the grid.

| abonoment_type | count(*) |
|----------------|----------|
| Premium        | 493      |
| Standard       | 507      |

**Insight: This shows which membership plan is the most popular.**

6.Find only one oldest member.

```
SELECT age, gender
```

```
FROM gym.gym_membership
```

```
ORDER BY age DESC
```

```
LIMIT 1;
```



The screenshot shows a database interface with a 'Result Grid' tab. The grid contains two columns: 'age' and 'gender'. There is one row: 49, Male. An 'Export' button, a 'Wrap Cell Content' checkbox, and a 'Fetch rows' button are visible at the top right.

| age | gender |
|-----|--------|
| 49  | Male   |

**Insight: This identifies the oldest member in the dataset.**

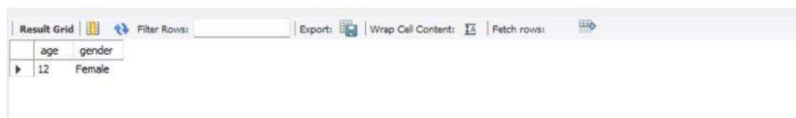
7.Find only one youngest member.

```
SELECT age, gender
```

```
FROM gym.gym_membership
```

```
ORDER BY age ASC
```

```
LIMIT 1;
```



The screenshot shows a database interface with a 'Result Grid' tab. The grid contains two columns: 'age' and 'gender'. There is one row: 12, Female. An 'Export' button, a 'Wrap Cell Content' checkbox, and a 'Fetch rows' button are visible at the top right.

| age | gender |
|-----|--------|
| 12  | Female |

**Insight: This identifies the youngest member in the dataset.**

8.Find the average age for each membership type.

## Gym Membership Data Analysis using MYSQL

```
SELECT abonoment_type, AVG(age)
```

```
FROM gym.gym_membership
```

```
GROUP BY abonoment_type;
```



A screenshot of a database result grid. The grid has two columns: 'abonoment\_type' and 'avg(age)'. There are two rows: 'Premium' with a value of 30.8641 and 'Standard' with a value of 30.3511. The interface includes a 'Filter Rows' button, an 'Export' button, and a 'Wrap Cell Contents' checkbox.

| abonoment_type | avg(age) |
|----------------|----------|
| Premium        | 30.8641  |
| Standard       | 30.3511  |

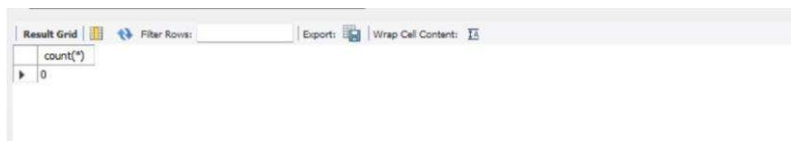
**Insight: This compares the average age of members across different membership plans.**

9.Count the number of sauna users.

```
SELECT COUNT(*)
```

```
FROM gym.gym_membership
```

```
WHERE uses_sauna = TRUE;
```



A screenshot of a database result grid. The grid has one column: 'count(\*)'. There is one row with a value of 0. The interface includes a 'Filter Rows' button, an 'Export' button, and a 'Wrap Cell Contents' checkbox.

| count(*) |
|----------|
| 0        |

**Insight: This shows how many members use the sauna facility.**

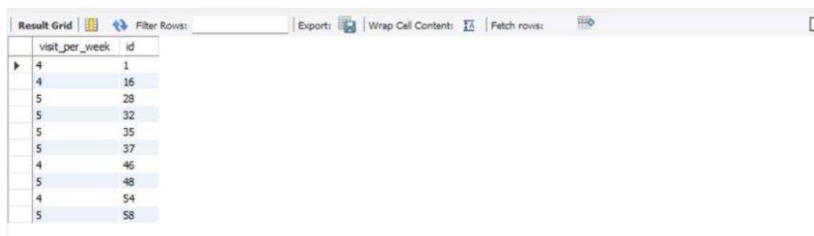
10.Find only 10 members who visit the gym more than 3 times per week.

```
SELECT visit_per_week, id
```

```
FROM gym.gym_membership
```

```
WHERE visit_per_week > 3
```

```
Limit 10;
```



A screenshot of a database result grid. The grid has two columns: 'visit\_per\_week' and 'id'. There are 10 rows. The values for 'visit\_per\_week' are 4, 4, 5, 5, 5, 5, 4, 5, 4, 5. The values for 'id' are 1, 16, 28, 32, 35, 37, 46, 48, 54, 58. The interface includes a 'Filter Rows' button, an 'Export' button, a 'Wrap Cell Contents' checkbox, and a 'Fetch rows' button.

| visit_per_week | id |
|----------------|----|
| 4              | 1  |
| 4              | 16 |
| 5              | 28 |
| 5              | 32 |
| 5              | 35 |
| 5              | 37 |
| 4              | 46 |
| 5              | 48 |
| 4              | 54 |
| 5              | 58 |

**Insight: This identifies highly active gym members.**

## Gym Membership Data Analysis using MYSQL

11.Display only 10 premium members sorted by age.

```
SELECT abonoment_type, age  
FROM gym.gym_membership  
WHERE abonoment_type = 'premium'  
ORDER BY age  
LIMIT 10;
```



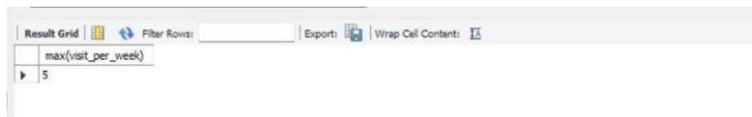
The screenshot shows a database query result grid with two columns: 'abonoment\_type' and 'age'. The results are sorted by age in ascending order. The first 9 rows show members with age 12, and the 10th row shows a member with age 13. All members are of type 'Premium'.

| abonoment_type | age |
|----------------|-----|
| Premium        | 12  |
| Premium        | 12  |
| Premium        | 12  |
| Premium        | 12  |
| Premium        | 12  |
| Premium        | 12  |
| Premium        | 12  |
| Premium        | 12  |
| Premium        | 12  |
| Premium        | 13  |

**Insight:** This helps analyze the age distribution of premium members.

12.Find the maximum number of visits per week.

```
SELECT MAX(visit_per_week)  
FROM gym.gym_membership;
```



The screenshot shows a database query result grid with one column: 'max(visit\_per\_week)'. The result is a single value, 5.

| max(visit_per_week) |
|---------------------|
| 5                   |

**Insight:** This shows the highest gym visit frequency among members.

13.Count members for each favorite drink.

```
SELECT fav_drink, count(*)  
FROM gym.gym_membership  
GROUP BY fav_drink;
```



The screenshot shows a database query result grid with two columns: 'fav\_drink' and 'count(\*)'. The results are grouped by favorite drink. The most popular drink is 'berry\_boost, lemon' with 504 members. Other popular drinks include 'passion\_fruit' (52), 'coconut\_pineapple' (53), and 'lemon' (47).

| fav_drink             | count(*) |
|-----------------------|----------|
| berry_boost, lemon    | 504      |
| passion_fruit         | 52       |
| coconut_pineapple     | 53       |
| orange, lemon         | 6        |
| black_currant         | 39       |
| orange, black_currant | 12       |
| orange                | 40       |
| lemon                 | 47       |
| passion_fruit, orange | 12       |

**Insight: This identifies the most preferred drink among gym members.**

14. Find only 10 members aged between 25 and 40.

```
SELECT age, gender
```

```
FROM gym.gym_membership
```

```
WHERE age BETWEEN 25 AND 40
```

```
LIMIT 10;
```



| age | gender |
|-----|--------|
| 27  | Female |
| 30  | Male   |
| 27  | Male   |
| 30  | Female |
| 35  | Female |
| 35  | Female |
| 36  | Female |
| 37  | Male   |
| 38  | Female |
| 33  | Male   |

15. Display only 10 members who attend on Monday.

```
SELECT days_per_week, visit_per_week
```

```
FROM gym.gym_membership
```

```
WHERE days_per_week LIKE '%MON%'
```

```
LIMIT 10;
```



| days_per_week      | visit_per_week |
|--------------------|----------------|
| Mon, Sat, Tue, Wed | 4              |
| Mon, Sat, Wed      | 3              |
| Mon                | 1              |
| Mon, Wed           | 2              |
| Mon                | 1              |
| Mon, Tue, Wed      | 3              |
| Mon, Thu, Tue, Wed | 4              |
| Fri, Mon           | 2              |
| Fri, Mon, Sat      | 3              |
| Mon, Tue           | 2              |

**Insight: This identifies members who regularly attend on Mondays.**

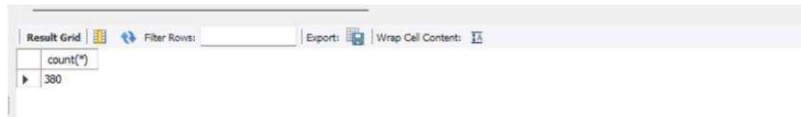
16. Count members attending on Saturday.

```
SELECT COUNT(*)
```

```
FROM gym.gym_membership
```

```
WHERE days_per_week LIKE '%sat%';
```

## Gym Membership Data Analysis using MYSQL



The screenshot shows a MySQL query result grid. At the top, there are tabs for 'Result Grid', 'Filter Rows', 'Export', and 'Wrap Cell Content'. Below these, a table with one row is displayed. The first column is labeled 'count(\*)' and the value in the cell is '380'.

| count(*) |
|----------|
| 380      |

**Insight: This shows the number of members who visit the gym on Saturdays.**

## CONCLUSION

This project helped me understand how to use MYSQL to analyze real world data. I learned SQL concepts such as **SELECT** , **WHERE**, **GROUP BY**, **ORDER BY**, **LIMIT**, Aggregate Functions (**COUNT**, **AVG**, **MIN**, **MAX**) and filtering. By analyzing the Gym Membership data, I was able to identify useful patterns in membership types, age groups, attendance and gym usage. This project improved my SQL querying and data analysis skills.

Overall this project improved my SQL skills, data analysis techniques and confidence in working with real data. It also gave me hands on experience in organizing, querying, and presenting data effectively.

